Imperial College London

Department of Earth Science and Engineering

MSc Individual Research Project

Causal Structure Discovery via Self-Implemented FCI$^+$ Algorithm:

A Case Study on the Tennessee STAR Project

Author: Yicheng Qi

Project code: p-anli-051

Main supervisor: Ang Li

Florida State University | angli@cs.fsu.edu

Second supervisor: Adriana Paluszny

Imperial College London | apaluszn@imperial.ac.uk

IRP repository:

https://github.com/ese-ada-lovelace-2025/irp-yq225

August 2026

Abstract

This project independently implements the FCI and FCI+ causal-discovery algorithms, validates them against exact graphical oracles and established software, and applies them to the Tennessee STAR class-size experiment. FCI learns partial ancestral graphs (PAGs) that remain valid under latent confounding and selection, but its Possible-D-SEP search is expensive; FCI+ attains a polynomial conditional-independence (CI) query bound for sparse graphs. Both search schedules are implemented in a self-contained, fully typed Python package (9,559 lines, 38 modules, no causal-discovery dependencies), validated by 367 automated tests, exact m-separation fixtures, a 15-run known-truth benchmark, and differential comparisons with causal-learn and R `pcalg`. On the benchmark, a robust application profile of FCI+ attains mean skeleton F1 0.985 and endpoint accuracy 0.840, against 0.977 and 0.737 for paper-profile FCI. The separate STAR application (6,325 kindergarten-cohort students) finds a design-supported small-class advantage of 13.90 points in observed kindergarten achievement (school-cluster 95% interval 5.21 to 22.07) and near-zero aide contrasts; because the outcome subset is post-assignment selected, this is not an identified causal effect. Across methods, skeleton-level structure proves more defensible than several order-, threshold-, or chronology-sensitive endpoint marks; no PAG here identifies a unique DAG or estimates a treatment effect.

Keywords: causal discovery; FCI; FCI+; partial ancestral graph; latent confounding; Tennessee STAR.

1 Introduction

1.1 Problem description

Many empirical studies ask causal questions where not every common cause has been measured: family background, institutional selection, and administrative recording generate dependence that no causally sufficient DAG over the recorded columns represents honestly, and fitting one anyway turns uncertainty into unjustified arrows. FCI was designed for this setting: under causal Markov, faithfulness, and appropriate sampling assumptions, it learns from conditional-independence (CI) information a Partial Ancestral Graph (PAG) that remains valid under latent confounding and selection (Spirtes et al., 2000; Richardson and Spirtes, 2002; Glymour et al., 2019), at the price of an expensive Possible-D-SEP search and endpoint ambiguity; Claassen et al. (2013) prove a polynomial worst-case CI-query bound for sparse graphs of fixed maximum degree k.

The research problem is twofold: can FCI and FCI+ be reimplemented faithfully and verifiably from their primary sources, and what can such an implementation legitimately conclude from a real randomized educational study — the Tennessee STAR class-size experiment — whose observed data are shaped by attrition and selection?

1.2 Significance

Public FCI-family tools typically return a finished graph with little visibility into why each endpoint was oriented; among the interfaces checked in section 5, no Python implementation of Claassen's FCI+ with paper-profile control and a structured audit trail was identified. An implementation whose every mark traces to recorded CI decisions serves applied researchers defending structural claims, methodologists needing a typed reference for differential testing, and instructors separating design-based from discovery evidence. The application adds substantive interest: STAR remains a canonical evidence source on class size, and selective attrition in longitudinal experiments is a live methodological concern.

1.3 Objectives and contributions

This project independently implements paper-aligned FCI and FCI+, validates them against exact graphical oracles and established software, and separates algorithm fidelity from a robust finite-sample application workflow, delivered as four auditable contributions.

C1. *Algorithm implementation.* A self-contained, fully typed Python package (`fci_engine`) written from

the primary sources without any causal-discovery dependency: five CI tests, a PAG endpoint algebra, the ordered Possible-D-SEP and complete Algorithm 2 stages, and five orientation schedules.

C2. *Validation apparatus.* An exact MAG m-separation oracle, encoded published fixtures, a 15-run known-truth benchmark, and 367 automated tests under strict typing on Python 3.9–3.13.

C3. *Software positioning.* Executed differential comparisons against causal-learn and R `pcalg` under matched CI semantics, documenting the integration gap identified above.

C4. *Application.* A separate Tennessee STAR case study, plus an independent R replication runner, consuming only the public API, with school-cluster bootstrap, cyclic-order, chronology, and sensitivity audits under a pre-specified evidence hierarchy.

C1 and C4 are deliberately separated: the algorithm stays reusable and domain-neutral, while the application states its own coding, selection, clustering, and interpretation decisions.

Five questions guided the work. **RQ1**: how do the sources define their targets, assumptions, and guarantee boundaries? **RQ2**: does the implementation map those stages to explicit code and tests? **RQ3**: what is recovered under exact-oracle and finite-sample known-truth conditions? **RQ4**: how does the package compare with established FCI software? **RQ5**: which STAR conclusions are design based, robust at the skeleton level, unresolved at the endpoint level, or merely exploratory? Each has a measurable criterion: the concept-to-symbol-to-test mapping (RQ1–RQ2), oracle-fixture recovery and benchmark metrics (RQ3), matched-semantics comparisons (RQ4), and the pre-specified promotion rules of section 6 (RQ5). Sections 2 and 3 address theory and implementation, sections 4 and 5 controlled evidence, and sections 6 to 8 the application; this ordering prevents a real-data pattern from serving as evidence that the algorithm matches its sources.

2 Background and Review of Existing Work

2.1 Graphs, assumptions, and PAGs

Constraint-based discovery begins from statements $X \perp Y \mid Z$ over observed variables V, with latent variables L and selection variables S possibly present. The *causal Markov condition* states that graphical separation implies conditional independence; *faithfulness* states that observed independences arise from separation rather than exact cancellation. Under causal sufficiency these support DAG methods such as PC. Otherwise the appropriate observed-variable object is a maximal ancestral graph (MAG) (Richardson and Spirtes, 2002): marginalising L and conditioning on S induce directed, bidirected, and undirected relations, and every missing MAG edge corresponds to an m-separation. Because different MAGs can encode the same independence model, the identifiable output is a PAG, recording exactly the endpoint features shared by one Markov equivalence class (Zhang, 2008).

Each PAG adjacency carries two endpoint marks: a circle leaves ancestry unresolved, an arrowhead at Y excludes Y as an ancestor of X, a tail asserts ancestry in every represented MAG, a bidirected edge is compatible with latent confounding without identifying a unique latent variable, and undirected edges reflect selection structure. An adjacency says only that the tested sets failed to separate the pair; neither an adjacency nor an endpoint alone identifies a structural coefficient, an adjustment set, or an interventional contrast (Pearl, 2009). M-separation generalises d-separation and is the criterion used by this project's exact oracle; sample tests are error-prone, so oracle recovery isolates graph logic rather than finite-sample accuracy. Among the discovery families reviewed by Glymour et al. (2019), score-based search (such as GES) and standard functional causal models (such as LiNGAM) assume causal sufficiency, so the constraint-based FCI family is the appropriate class when latent confounding and selection must be represented.

2.2 Standard FCI and its historical boundary

FCI does not require that all common causes of measured variables are measured (Spirtes et al., 2000, Ch. 6). A PC-style search removes $X - Y$ when some subset of current neighbours separates the pair,

but with latent variables a separable pair may stay adjacent: a necessary separator can contain a node no longer adjacent to either endpoint. The book introduces D-SEP to characterise where an observed separator can reside and, since D-SEP is not computable from a partially learned graph, the observable superset Possible-D-SEP: for an ordered pair (A, B), a node $V \neq A$ qualifies when some path from A to V has every internal triple either a collider or a triangle whose middle node is not a definite noncollider. The construction is deliberately permissive and required *separately* from each ordered endpoint: using one direction only, or mixing pools, changes the search contract. The full schedule: PC-style adjacency search; initial unshielded colliders; ordered Possible-D-SEP refinement; endpoint reset and collider reconstruction from final separators; orientation closure. Theorem 6.4 guarantees a correct *partially oriented inducing path graph* under correct population-level CI decisions — while the authors state they did not know whether the output was maximally informative. Complete orientation therefore depends on the later rules of Zhang (2008) and is not retroactively attributed to the 2000 source.

2.3 FCI$^+$: Algorithm 2

Claassen et al. (2013) ask whether causal discovery remains intrinsically exponential when the observed structure is sparse. Their task assumes a constant-time exact independence oracle, faithfulness to a causal DAG, an observed-variable MAG of maximum degree at most a constant k, and a completed PAG as output. Under those assumptions they prove the worst-case CI-query bound $O(N^{2(k+2)})$ for N observed variables — narrower than the paper's title: it bounds oracle queries and does not make a statistical test exact or constant-time.

A nonadjacent pair in the true MAG can survive bounded adjacency search when every useful separator requires a node outside the endpoint-adjacency union; such a false adjacency is a *D-sep link*. Rather than searching a large Possible-D-SEP pool, FCI$^+$ extracts invariant non-ancestry into an *augmented skeleton* G^+: when a known minimal independence $X \perp Y \mid S$ becomes dependence given $S \cup \{A\}$ for a suitable adjacent A, an invariant arrowhead at A is supported. In G^+, a D-sep link $X - Y$ appears as the middle edge of $U \leftrightarrow X \leftrightarrow Y \leftrightarrow V$ with U, V nonadjacent and the required cross-direction paths not running against an arrowhead; the pattern only *proposes* a candidate, and independence evidence decides deletion. Conditioning sets come from the recursive hierarchy $\mathrm{HIE}(X, I)$, the fixed point of repeatedly adding recorded minimal separators of members; because candidates can depend on one another, each successful removal rebuilds the augmented skeleton and candidate list (figure 1). The arXiv proof supplement's abstract states $N^{2(k+1)}$ while its theorem and proof agree with the UAI paper's $N^{2(k+2)}$, a discrepancy recorded rather than harmonised. Algorithm 2 invokes complete FCI orientation rather than rederiving it, so its completed-output claim inherits Zhang (2008) as a dependency.

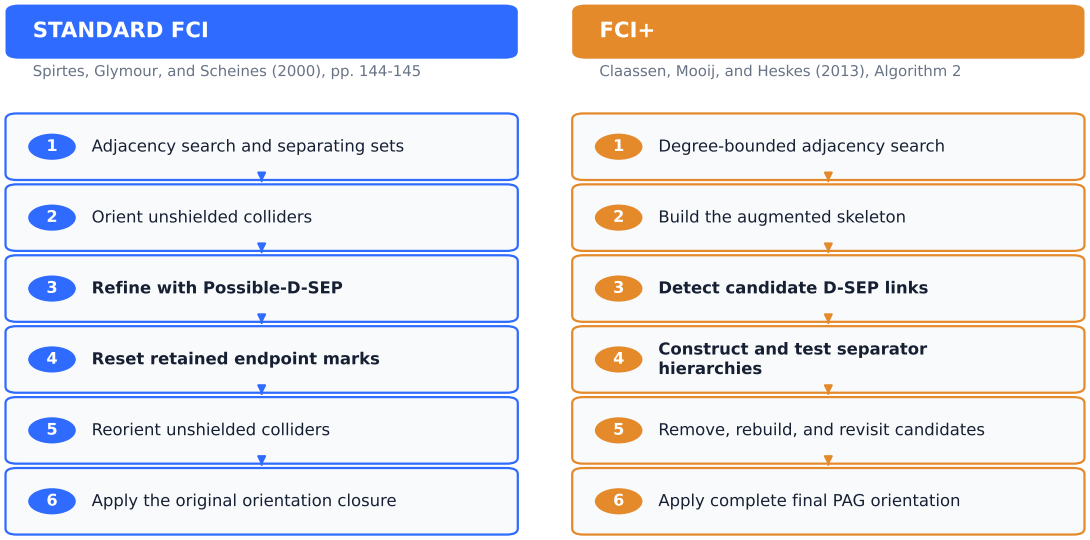


Figure 1: Source-located control-flow comparison. Standard FCI performs a non-local Possible-D-SEP refinement; FCI⁺ instead iterates augmented-skeleton, candidate D-sep-link, hierarchy, and minimal-separator stages before invoking final orientation. The robust application profile of section 3 changes finite-sample policy, not the definition of Algorithm 2.

2.4 What the sources do not establish

Neither source guarantees finite-sample recovery: both theorems are conditional on correct CI decisions. Neither identifies a unique DAG, an adjustment set, or a treatment effect; a bidirected pattern does not name or measure a hidden cause. The 2000 source proves no maximal orientation and no polynomial bound; the 2013 source proves no finite-sample superiority and nothing when k is wrong — a user-entered bound is not evidence — and its authors warn that hierarchy-derived conditioning sets may be large, reducing finite-sample power.

3 Independent Python Implementation

3.1 Package boundary and public API

The package exposes the algorithmic obligations of section 2 as inspectable components, separate from finite-sample application policy. Reusable code resides in `src/fci_engine`, the STAR workflow in `case_studies/tennessee_star`; no STAR-specific choice is embedded in the estimator.

The algorithms are separate entry points with explicit paper profiles, so a finite-sample default cannot silently stand in for the source algorithm. `FCIConfig.paper()` fixes unbounded depth, immediate deletion, first-found separators, enabled Possible-D-SEP, nonconservative colliders, and the historical `spirtes_2000` closure; `FCIPlusConfig.paper(k)` requires an explicit k for both conditioning depth and sparsity bound, disables Possible-D-SEP, and selects the complete `standard` closure required by Algorithm 2's output claim. Profiles constrain execution; they do not validate theorem assumptions — supplying $k = 3$ does not prove the true MAG has degree at most three.

The `PAG` class stores an endpoint mark at each side of every adjacency, keeping tails, arrowheads, circles, and absence explicit. Every search uses one `CITest.test(data, x, y, cond_set)` contract, with built-in Fisher-Z, missing-value Fisher-Z, chi-square, G-square, and kernel tests. A cache canonicalises each query, so results separate the *logical* CI requests made by control flow from cache hits and executed statistics — the distinction relevant to a theorem that counts oracle queries.

3.2 Pipelines, orientation schedules, and result objects

`FCI.fit` exposes the source stage ordering directly, recording Possible-D-SEP removals with source `pdsep`. `FCIPlus.fit` deliberately does not call Possible-D-SEP; it passes the initial skeleton to the Algorithm 2 search: augmented skeleton, bidirected witness patterns, paired endpoint bases, hierarchy fixed points, separator testing and minimisation, graph update with source `fci_plus_dsep`, and candidate rebuild. A diagnostics record counts candidates, revisits, hierarchy queries, D-SEP-stage CI requests, and the maximum conditioning size — empirical accounting, not a mechanised proof of the bound.

One dispatcher provides five orientation schedules: `spirtes_2000`, the historical closure used by the standard-FCI paper profile; `standard`, the modern complete schedule required by the FCI+ paper profile; and `conservative`, `leaf`, and `robust`, finite-sample engineering policies supporting no completed-PAG claim. A dedicated regression verifies that `spirtes_2000` stops before a selection-bias tail completion that `standard` can orient, making the historical completeness boundary executable rather than descriptive.

`FCIResult` binds the final graph to the evidence that produced it — configuration, separators with stage sources, logical CI counts, and named CI and orientation traces; `explain_edge` returns every decision behind a pair. A plotted arrow cannot become detached from the profile, test, threshold, and rule that created it.

3.3 Engineering scope and application policy

The code base was written entirely for this project: 9,559 lines across 38 typed modules in the algorithm package; 367 tests in 44 modules (5,929 lines); a 3,052-line STAR application plus a 396-line independent R runner; and benchmark and figure scripts, totalling roughly 25,300 lines. The package ships `py.typed` and passes Ruff, strict MyPy, a wheel build, and Pytest on Python 3.9–3.13; a full concept-to-symbol-to-test traceability matrix is maintained in the repository's extended report. These checks establish interface and packaging consistency, not faithfulness, true sparsity, or any empirical causal conclusion.

The practical profile trades computation for stability: stable deletion reduces order dependence; max-p separators spend more CI queries for stronger independence evidence; conservative collider and tail policies preserve circles; automatic α is a documented sample-size heuristic. Accordingly, "paper FCI+" is the fidelity and query-efficiency subject, and "robust FCI+" is the preferred policy for the reported application because it meets the stated audit criteria while returning less-specific endpoints — a local preference, not a universal ranking.

4 Algorithm Validation

4.1 Exact oracle tests

The strongest tests replace sample CI decisions with exact m-separation in a known MAG, isolating search correctness from sampling error. Three independently specified targets come from published structures: the canonical Figure 4(b) D-SEP example of Claassen et al. (2013), a latent-variable reference graph from the FCI literature, and an orientation-rule reference graph associated with Zhang (2008). Both algorithms recover all three targets, with Figure 4(b) additionally tested under four variable orders.

The Figure 4(b) MAG has six edges ($Z \to U$, $Z \to V$, $U \leftrightarrow X$, $V \to X$, $U \to Y$, $V \leftrightarrow Y$); the adjacency stage retains a false $X - Y$ link although $X \perp Y \mid \{U, V, Z\}$, with Z adjacent to neither endpoint. The source exhibits this separator but prints no endpoint-complete PAG, so the completed comparison target is repository derived (figure 2). On this example, paper-aligned standard FCI needs 102 CI queries and FCI+ 63 (a 38.2% reduction), both returning the exact target; the values are frozen in `reports/data/oracle_validation_summary.csv` with the target hash and separator provenance.

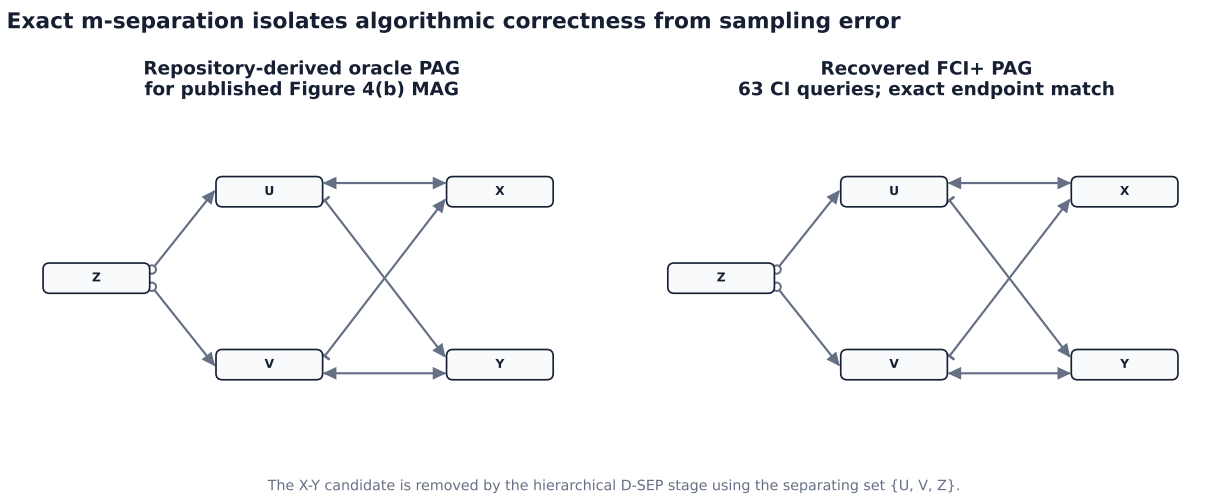


Figure 2: Exact-oracle validation on the central Figure 4(b) example. The source-grounded facts are the six MAG edges and the exhibited separator $\{U, V, Z\}$; the endpoint-complete comparison target is independently derived in the repository. Paper-aligned FCI needs 102 oracle queries, FCI+ 63; both recover the target exactly.

4.2 Finite-sample behaviour

On a seeded latent linear SEM over the same reference graph, at $N = 5{,}000$ FCI+ returns an extra $X \leftrightarrow Y$ (exact-edge F1 0.923) while standard FCI recovers the exact PAG; at $N = 50{,}000$ both are exact. This is a reproducible fixture, not a universal sample-size threshold.

The application safeguard was tested on 15 deterministic latent-SEM cases — five graph families, three seeds each, 2,500 observations, known oracle PAGs (table 1).

Table 1: Mean recovery across 15 finite-sample oracle cases.

Profile	Skeleton F1	Exact F1	Semantic F1	Endpoint acc.	CI calls
Paper FCI+	0.979	0.536	0.951	0.744	197.1
Robust FCI+	0.985	0.730	0.985	0.840	401.9

The robust profile improves the mean skeleton and endpoint scores at about 2.04 times the CI cost — not proof of universal dominance, but a known-truth check that the STAR stability gain is not obtained by discarding structure, and a warning that fewer CI tests do not imply better accuracy: the two algorithms take different logical routes after the initial skeleton and can make different errors.

5 Comparison with Established Software

The comparison separates execution from interface inspection: the executed layer asks what happened on identical committed data under recorded settings; the capability layer asks what each public interface or documentation exposes. Tetrad 7.6.10 (Tetrad Project, 2026) appears only in the latter because no working Java runtime was available; all claims are dated 26 July 2026.

The benchmark reuses the 15 known-truth runs, every method receiving the same numeric matrix and variable order. Fixed paper and reference rows use Fisher-Z at $\alpha = 0.001$; robust FCI+ uses its documented sample-size rule, which resolves to $\alpha = 0.01$ here — intentional: that row evaluates the public robust policy, not a literal Algorithm 2 reproduction (table 2).

Table 2: Executed known-truth comparison: means over 15 completed runs.

Method (version)	Skeleton F1	Exact F1	Semantic F1	Endpoint acc.	Seconds	Logical CI
Local FCI paper (0.1.0)	0.977	0.532	0.943	0.737	0.020	449.7
Local FCI⁺ paper (0.1.0)	0.979	0.536	0.951	0.744	0.009	197.1
Local FCI⁺ robust (0.1.0)	0.985	0.730	0.985	0.840	0.018	401.9
causal-learn FCI (0.1.4.7)	0.979	0.536	0.951	0.744	0.009	not exposed
`pcalg::fciPlus` (2.7-12)	0.979	0.536	0.951	0.744	0.500	not exposed

Exact F1 requires both endpoint marks to match; semantic F1 additionally credits endpoint patterns compatible with the target ancestral constraints; endpoint accuracy scores individual marks. The common conversion represents each edge as an unordered pair plus two semantic marks, never as generic directed edges. The equality of the aggregate local paper-FCI⁺, causal-learn (causal-learn developers, 2026), and `pcalg` (Kalisch et al., 2012) scores is useful differential evidence, not an equivalence proof: local and causal-learn times cover in-process calls whereas the R timing includes process startup, and `pcalg::fciPlus` exposes no identical control for the degree bound k. Throughout, data, order, alpha, and CI semantics are held fixed where interfaces allow, every mismatch is disclosed, and distinct FCI-family algorithms are never relabelled as one.

On the capability layer, `pcalg` exposes both algorithms and accepts user-defined CI tests, but its `fciPlus` result carries materially less separator provenance; causal-learn exposes standard FCI but no public Claassen FCI⁺ entry point; Tetrad provides FCI-family hybrids, none established as a drop-in Claassen FCI⁺. The local package integrates paper-profile control, structured traces, separator provenance, per-edge explanations, and stability auditing in one workflow — an auditability improvement, not a universal accuracy or speed claim.

6 STAR Study Design and Methods

6.1 Experiment, cohort, and panels

The Tennessee Student/Teacher Achievement Ratio project was a large randomized experiment in kindergarten through grade 3, assigning students and teachers to small classes (about 13–17 students), regular classes (about 22–25), or regular classes with a full-time aide (Word et al., 1990); it remains a central source of evidence on class size (Krueger, 1999). This analysis uses the official Harvard Dataverse release (Achilles et al., 2008): 11,601 student records, 379 variables, CC0 licence, with codes checked against the accompanying 146-page user guide.

Randomization already provides a stronger basis for estimating the class-size effect than structure learning, so FCI is not used to rediscover it. The purposes are: whether grade-3 observation relates structurally to early achievement or background; whether early and later achievement remain adjacent given context; how the PAG changes when an early post-assignment outcome is included or excluded; whether implementations agree; and how much search work FCI⁺ saves. The design also supplies an external audit: a data-only PAG implying an implausible pre-treatment direction, or latent confounding of a randomized assignment, should be reported as a discrepancy rather than accepted literally.

The analysis covers the 6,325 students with a kindergarten class assignment across 79 schools, arranged into three panels deliberately not pooled because each encodes a different statistical question (table 3). All model inputs are integer-coded discrete categories: class arm, sex, ethnicity (small source categories collapsed), free-lunch status, school context, entry-age quantiles, binned teacher experience, and reading-plus-mathematics scaled scores converted to achievement quartiles; the exact numeric tables supplied to the algorithms are committed in the repository.

Table 3: STAR analysis panels.

Panel	Rows	Nodes	Purpose
Attrition / observation	5,744	9	Relate assignment, characteristics, context, and kindergarten achievement to whether both grade-3 scores are observed.
Longitudinal achievement	2,787	9	Study structure between kindergarten and grade-3 achievement among complete cases while retaining early achievement.
Focused treatment-outcome	2,976	8	Study the class-assignment/grade-3 relationship without allowing kindergarten achievement, an early post-assignment outcome, to serve as a separator.

6.2 Analysis protocol

Before discovery, randomized-arm contrasts are computed for kindergarten scores, grade-3 scores among students with both scores, and the probability that both grade-3 scores are observed, with uncertainty from 1,000 kindergarten-school bootstrap samples preserving within-school dependence; these design-based contrasts are reported separately from discovery output.

Discovery uses the likelihood-ratio G-square test, $G^2 = 2\sum_s \sum_{x,y} O_{xys} \log(O_{xys}/E_{xys})$, at $\alpha = 0.05$ — chosen because every panel variable is an integer-coded category, where Fisher-Z's Gaussian-linear assumptions fail, at the cost that high-order conditioning can create sparse cells. Standard FCI runs its unbounded paper schedule; paper FCI$^+$ uses $k = 3$; robust FCI$^+$ uses the practical profile (stable skeleton, strongest separating sets, conservative colliders, robust orientation, depth and sparsity bound 3); the independent reference is `pcalg::fciPlus` with selection bias enabled and no exposed k. No temporal priors are supplied to discovery; reversals are audited after fitting.

Three audits probe stability: 100 school-cluster bootstrap refits per algorithm and panel, recording each unordered adjacency's appearance frequency (an exploratory diagnostic, not a posterior probability); a cyclic-order audit refitting every cyclic shift of the variable list for all profiles and the R reference; and a focused-panel sensitivity grid over $\alpha \in \{0.01, 0.05\}$, three versus four quantile bins, and $k \in \{2, 3, 4\}$.

The application distinguishes a reproducible adjacency from a credible direction: a skeleton-level conclusion is promoted only when the adjacency survives every cyclic shift with high bootstrap frequency; a directional conclusion additionally requires invariant endpoint marks, no conflict with measurement time or randomization, and no material reversal under sensitivity checks. Failure downgrades the claim to unresolved direction rather than deleting the edge.

7 STAR Results

7.1 Observed randomized-arm outcomes

The small-minus-regular kindergarten contrast is 13.90 points (school-cluster 95% interval 5.21 to 22.07), while the regular-with-aide contrast is 0.31 points (−7.40 to 7.38). At grade 3 among observed students the small-class contrast remains 11.69 points (3.74 to 19.38) and the aide contrast 0.51 (−5.70 to 7.22). Observation rates show no precise arm difference: small minus regular is +1.54 percentage points (−1.73 to 4.56), aide minus regular −2.05 (−5.35 to 1.41) (table 4).

Table 4: Observed outcome summaries by kindergarten class arm. Intervals use 1,000 kindergarten-school bootstrap samples. Grade-3 summaries condition on both scores being observed; kindergarten summaries also exclude missing score components, so both are design-supported observed-outcome comparisons rather than a fully reconstructed intention-to-treat estimand.

Outcome	Arm	*n*	Estimate	95% interval
Kindergarten score	Regular	2,005	918.04	908.78 to 927.05
	Regular + aide	2,043	918.36	908.52 to 927.07
	Small	1,738	931.94	922.70 to 940.12
Grade-3 score	Regular	1,047	1246.74	1238.96 to 1254.01
	Regular + aide	1,019	1247.25	1238.53 to 1255.48
	Small	936	1258.43	1250.58 to 1266.17
Grade-3 observed	Regular	2,194	47.72%	42.72% to 52.14%
	Regular + aide	2,231	45.67%	40.64% to 50.39%
	Small	1,900	49.26%	44.46% to 53.77%

7.2 Attrition and longitudinal panels

On the attrition panel the three implementations substantially agree at the skeleton level (figure 3; costs and agreement in table 5). All three omit a direct `K_Class`–`Grade3_Observed` adjacency, which appears in only 12–16% of school bootstraps. By contrast, `K_Achievement`–`Grade3_Observed` is adjacent in 100% of bootstrap fits for all three local profiles and the free-lunch/observation adjacency is also stable: later availability is not independent of early achievement and measured context. The three endpoint readings diverge: standard FCI gives the impossible *grade-3 observation → kindergarten achievement*; self-FCI⁺ leaves a circle; R-FCI⁺ gives the temporally admissible direction. The R reading is more plausible, but package identity cannot establish statistical correctness. The attrition PAG also does not prove a missingness mechanism; a formal analysis would require an explicit estimand and additional assumptions.

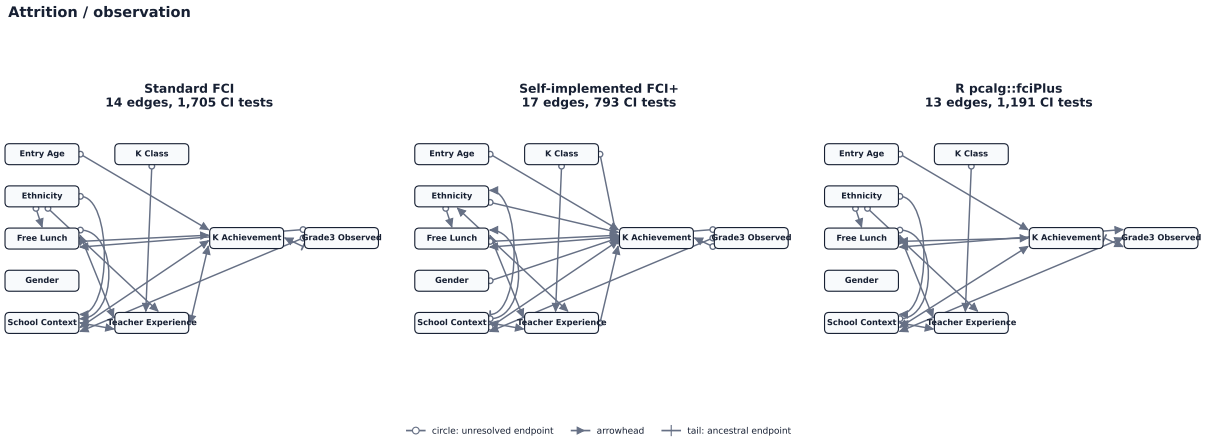


Figure 3: Three PAGs for the attrition/observation panel: self-implemented FCI, self-implemented FCI⁺, and R `pcalg::fciPlus`. The skeletons substantially overlap, while the achievement/observation endpoint differs across implementations.

On the longitudinal panel, all three implementations retain the kindergarten/grade-3 achievement adjacency, present in all 100 bootstrap samples for all three local profiles. No full-data PAG retains a direct class-assignment/grade-3 adjacency once kindergarten achievement is included, yet that adjacency reappears in 71% of FCI, 79% of paper FCI⁺, and 80% of robust-profile bootstrap fits — a warning that conservative orientation and order invariance do not remove school-resampling sensitivity. Both FCI⁺ implementations

orient grade-3 achievement toward kindergarten achievement, which is temporally impossible if read causally, while standard FCI leaves that endpoint unresolved. Agreement between implementations is not sufficient to override chronology: both may replicate the same finite-sample CI pattern under complete-case selection.

7.3 Focused treatment-outcome panel

Removing kindergarten achievement changes the result: all three algorithms return the same 10-edge PAG — not merely the same skeleton — including *K_Class* ↔ *Grade3_Achievement* (figure 4), an adjacency appearing in 92%, 85%, and 96% of FCI, paper, and robust bootstraps. Independent exact replication makes this conditional-independence pattern more credible, but it does not make a literal latent-confounding reading credible: kindergarten class assignment was randomized, while the focused panel conditions on grade-3 outcomes being observed and omits early post-assignment achievement. Selection on observation, school-level dependence, omitted post-randomization variables, and CI-test error can all produce arrowheads at both endpoints.

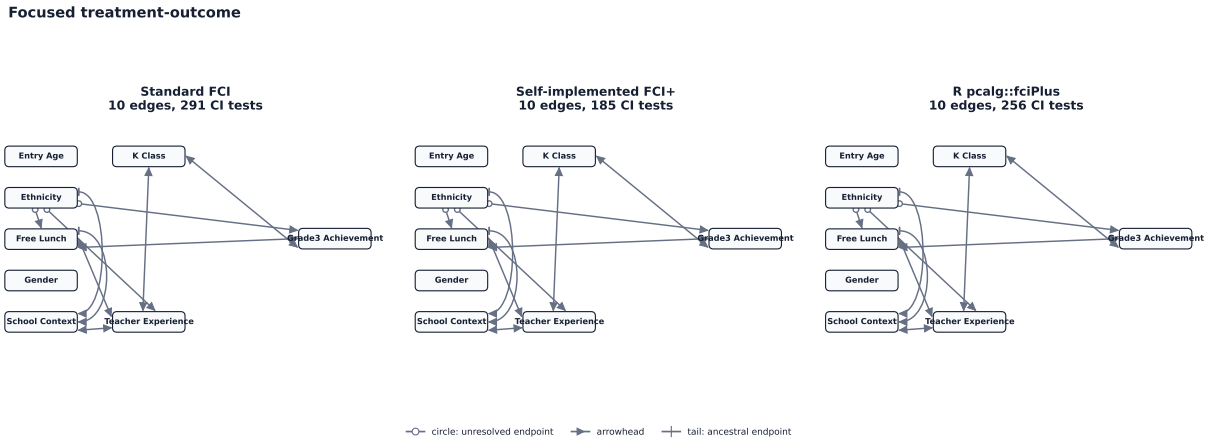


Figure 4: PAGs for the focused complete-case panel. All three algorithms recover the exact same 10-edge skeleton and endpoint pattern, including the bidirected class-assignment/grade-3 edge.

The sensitivity result is decisive (figure 5): the focused adjacency is robust to the tested bin count and degree bound but not to the CI threshold: at $\alpha = 0.01$ no local configuration retains it, while at $\alpha = 0.05$ the paper profiles retain a bidirected mark for every tested k and the robust endpoint keeps circles. The stricter threshold demands more evidence against independence, so removal indicates the edge rests on relatively weak conditional-dependence evidence rather than an invariant feature.

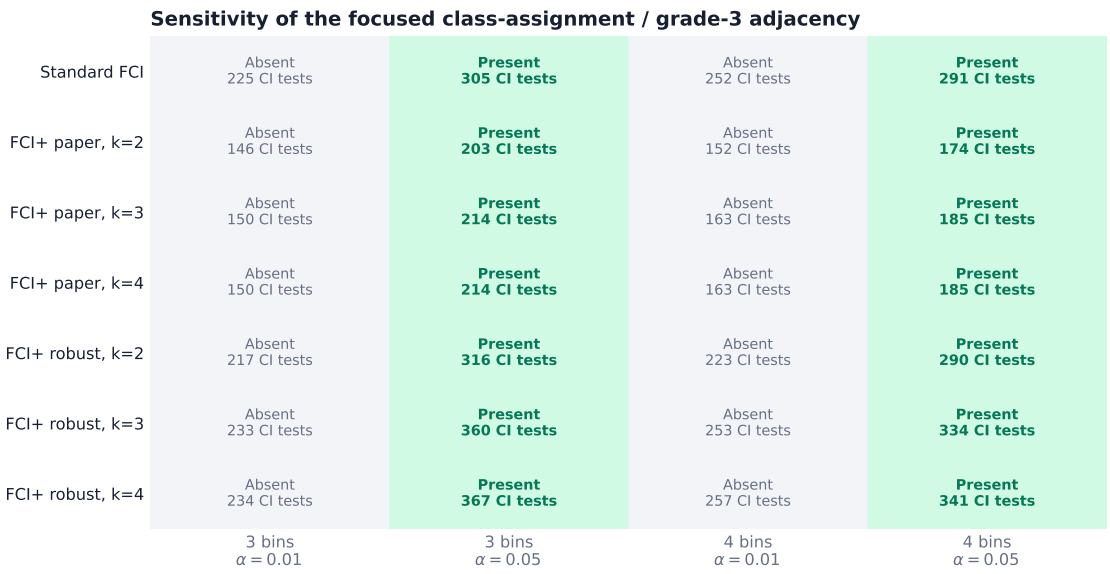


Figure 5: Focused treatment-outcome sensitivity. The adjacency appears at $\alpha = 0.05$ and disappears at $\alpha = 0.01$, regardless of whether three or four quantile bins or $k = 2, 3, 4$ is used for FCI⁺.

7.4 Robust application profile, agreement, and cost

The robust profile is a finite-sample decision layer applied after the literal comparisons, not a fourth paper implementation. It returns 17, 12, and 10 edges on the three panels, with target relations *K_Achievement* ∘–∘ *Grade3_Observed*, *K_Achievement* ∘–∘ *Grade3_Achievement*, and *K_Class* ∘–∘ *Grade3_Achievement*. It retains the relevant skeleton information without the paper-profile chronology reversal or the focused-panel bidirected over-interpretation.

In the cyclic-order audit, paper self-FCI⁺ reproduces the exact full-data PAG in 9 of 9 attrition, 5 of 9 longitudinal, and 6 of 8 focused shifts, with one longitudinal temporal reversal; the R reference preserves every panel skeleton while matching exact endpoints in 7 of 9, 6 of 9, and 6 of 8 shifts; robust self-FCI⁺ returns the identical PAG in all 26 refits with no reversal. Table 5 summarises agreement and search cost. Paper FCI⁺ uses 46.5%, 34.9%, and 63.6% of the standard-FCI CI-test counts on the three panels (R reference: 69.9%, 49.5%, 88.0%). Runtime is not language neutral, so CI-call counts are the more interpretable comparison. Relative to paper self-FCI⁺, the robust profile uses 1.64 to 1.85 times as many CI calls, purchasing stable deletion and invariant conservative endpoints rather than speed.

Table 5: Three-algorithm structural agreement (left) and search cost (right). Endpoint rates use only adjacencies shared by the indicated pair; time is the median of three full-data repetitions.

Panel	Pair	Jaccard	Endpoints
Attr.	FCI / self	0.824	50.0%
	FCI / R	0.929	61.5%
	self / R	0.765	38.5%
Long.	FCI / self	0.769	50.0%
	FCI / R	0.833	70.0%
	self / R	0.923	83.3%
Focus.	FCI / self	1.000	100.0%
	FCI / R	1.000	100.0%
	self / R	1.000	100.0%

Panel	Algorithm	Edges	CI	Sec.
Attr.	FCI	14	1,705	5.437
	self paper	17	793	1.788
	self robust	17	1,304	2.838
	R-FCI+	13	1,191	10.394
Long.	FCI	10	960	2.148
	self paper	13	335	0.337
	self robust	12	619	0.562
	R-FCI+	12	475	1.650
Focus.	FCI	10	291	0.322
	self paper	10	185	0.159
	self robust	10	334	0.247
	R-FCI+	10	256	0.884

Key result. The robust profile returns the same full PAG for every tested cyclic order, every target adjacency survives, and no directed edge runs backward in measurement time — at the price of under-orientation and additional CI work. Reliability has improved; identifiability has not.

8 Discussion: Evidence Hierarchy and Limitations

8.1 Evidence hierarchy

An explicit hierarchy prevents a visually definite PAG mark from outranking the experimental design. Tier 1: randomized assignment with the school-cluster contrast, permitting a bounded causal statement for the analysed observed outcome. Tier 2: an adjacency replicated across methods, resamples, orders, and sensitivity settings — a robust skeleton-level claim with no magnitude or direction. Tier 3: a stable skeleton with circles or endpoint disagreement, where adjacency is more credible than orientation. Tier 4: an implementation-, order-, or chronology-sensitive endpoint — diagnostic evidence only. Tier 5: a threshold-sensitive or complete-case exploratory pattern — a hypothesis for external validation.

8.2 What can and cannot be concluded

Among kindergarten-cohort students with observed score components, the small-class arm is 13.90 points above the regular arm (school-cluster 95% interval 5.21 to 22.07) — consistent with a beneficial early effect, but not an identified causal effect in the post-assignment observed subset without assumptions about missing outcomes, compliance, and the original randomization blocks. The aide contrasts are close to zero at both time points: no evidence that an aide reproduces the small-class advantage. Kindergarten and grade-3 achievement remain adjacent in all full longitudinal fits and in 100% of school bootstraps, and kindergarten achievement is adjacent to grade-3 observation in all attrition fits: a robust *skeleton-level* relationship between early performance, later performance, and later availability. The full-data attrition PAGs do not retain a direct class-arm/observation adjacency and its bootstrap frequency is low, supporting concern about selective attrition without a stable direct effect of arm on observation.

Conversely: no unique DAG is identified — every PAG represents an equivalence class — and no effect magnitude comes from FCI. A bidirected edge is not proof of a hidden confounder; in the focused panel it conflicts with the known randomization unless selection or post-treatment conditioning is considered. Temporal reversals are algorithm diagnostics, not credible chronology. Implementation agreement is not proof: shared CI decisions and rules can reproduce the same wrong endpoint. The 26 recorded shifts are a finite audit, not a proof of order invariance; bootstrap frequency is not a posterior probability. Absence is

policy conditional: failure to retain an edge under one node set, test, and threshold is not proof of zero effect.

8.3 Limitations

Achievement and age are reduced to quantile categories and teacher experience is binned, which enables G-square testing but discards within-bin information and can leave small expected counts as conditioning sets grow. The CI test treats student rows as independent; the school bootstrap measures sensitivity to school composition but does not make each test cluster-robust. The longitudinal and focused panels require grade-3 scores, so conditioning on complete cases may open selection paths — a concern reinforced by the attrition panel itself. Kindergarten achievement is measured during the treatment period, so including it may block part of the class-size pathway while excluding it may leave omitted post-treatment dependence; both specifications are reported rather than selecting the one with the preferred edge. FCI-family guarantees rely on faithfulness and correct CI decisions, and the threshold-sensitive focused edge is a direct example of that limitation. The true maximum degree of the STAR MAG is unobserved, so $k = 3$ is a bounded analysis setting rather than a verified assumption. The package is version 0.1.0; hand-authored oracle targets are strong regression evidence, not a formal proof or an exhaustive MAG-equivalence check.

A deterministic scaling audit (5 to 80 nodes; a directed chain and the Figure 4(b) core with isolated variables added) recovers the exact target skeleton in every run but yields an unexpected finding — the query advantage is not universal: FCI⁺ wins on the five-node core (62 versus 101 queries) yet spends 5,162 versus 3,251 at 80 nodes, and 24,338 versus 12,638 on the 80-node chain — consistent with a worst-case bound; augmentation has a real cost when targeted D-SEP search offers little benefit. Substantively, original randomization blocks, class switching, compliance, and school-specific assignment procedures are not reconstructed, and the composite score simply adds reading and mathematics, leaving subject-specific and subgroup effects unexamined.

9 Conclusions and Future Work

Source fidelity is explicit and testable. Standard FCI maps to the historical 2000 schedule, whose Theorem 6.4 is a correctness result under correct CI information, not a modern maximal-orientation theorem; FCI⁺ maps to the augmented skeleton and hierarchy of Algorithm 2, whose polynomial result remains the restricted $O(N^{2(k+2)})$ exact-oracle query bound. Validation is layered rather than circular: exact m-separation tests recover independently specified targets; the 15-run known-truth benchmark shows robust FCI⁺ improving endpoint metrics and paper FCI⁺ reducing queries; causal-learn and `pcalg` supply differential evidence. The package's defensible improvement is its integrated audit surface, not a claim to be universally fastest or most accurate.

Applied to STAR without mixing domain code into the algorithm, the workflow shows a clear observed small-class advantage, no comparable aide advantage, stable early-to-later achievement structure, and evidence that later observation is selective. An independent R run exactly replicates the focused 10-edge PAG while showing why PAGs require discipline: the bidirected treatment-outcome edge is conditional on complete cases and disappears under a stricter alpha, and some endpoint marks change with order or violate chronology. The robust profile addresses the principal failures found by those audits — identical output across every tested order, preserved target adjacencies, no directed temporal reversal — at the cost of unresolved endpoints and more CI calls.

The most challenging aspects were pinning down the sources' exact search contracts — the ordered Possible-D-SEP pools and Algorithm 2's hierarchy and revisit semantics — and diagnosing finite-sample endpoint instability. Oracle-first validation and the pre-specified audits worked well; literal paper profiles, though faithful, proved fragile on real panels, so full-data PAGs alone would have been misleading.

No algorithm is uniformly best on these data: paper self-FCI⁺ is the fidelity and query-efficiency subject; robust self-FCI⁺ is preferred for applied interpretation under the predeclared audits; R `pcalg` provides independent replication and the most plausible attrition direction yet shares the longitudinal reversal;

standard FCI is less economical but leaves that target unresolved rather than reversed. Credibility must be assigned claim by claim. The report rejects two extremes: too pessimistic to conclude that causal discovery adds nothing to a randomized study — the attrition and stability analysis reveals useful structure — and too optimistic to conclude that FCI has rediscovered the experiment's causal model. The contribution lies between: an auditable map of conditional-independence constraints, uncertainty, and modelling sensitivity.

Recommended extensions are to reconstruct the randomization strata and estimate formal intention-to-treat effects with an explicit model for grade-3 observation; develop cluster-aware CI tests; extend stability summaries to endpoint compatibility; analyse reading and mathematics separately; evaluate defensible temporal background knowledge separately from the data-only robust profile; and add an exhaustive MAG-equivalence oracle for target-PAG derivation.

Reproduction. The full STAR application is rebuilt by the `run_case_study` module of `case_studies.tennessee_star` with `--benchmark-repeats 3`, `--bootstraps 100`, and `--descriptive-bootstraps 1000`; `--refresh-pcalg` re-executes the R reference. The scripts under `reports/` regenerate every figure, the frozen Figure 4(b) evidence, and the scaling audit. The committed `STAR_Students.tab.gz` has SHA-256 `2ffa578822eb30fcd6626fa7c5cc734a721fd05dd522403f0d643f89e891bac8`.[1]

References

Achilles, C. M., Bain, H. P., Bellott, F., Boyd-Zaharias, J., Finn, J., Folger, J., Johnston, J., and Word, E. (2008). Tennessee's Student Teacher Achievement Ratio (STAR) project. Harvard Dataverse, V1. `https://doi.org/10.7910/DVN/SIWH9F`.

Claassen, T., Mooij, J. M., and Heskes, T. (2013). Learning sparse causal models is not NP-hard. In *Proceedings of the 29th Conference on Uncertainty in Artificial Intelligence*.

causal-learn developers (2026). *FCI: Fast Causal Inference*. Official documentation, accessed 26 July 2026.

Glymour, C., Zhang, K., and Spirtes, P. (2019). Review of causal discovery methods based on graphical models. *Frontiers in Genetics*, 10, 524.

Kalisch, M., Mächler, M., Colombo, D., Maathuis, M. H., and Bühlmann, P. (2012). Causal inference using graphical models with the R package pcalg. *Journal of Statistical Software*, 47(11), 1–26.

Krueger, A. B. (1999). Experimental estimates of education production functions. *The Quarterly Journal of Economics*, 114(2), 497–532.

Pearl, J. (2009). *Causality: Models, Reasoning, and Inference*, second edition. Cambridge University Press.

Richardson, T. S. and Spirtes, P. (2002). Ancestral graph Markov models. *The Annals of Statistics*, 30(4), 962–1030.

Spirtes, P., Glymour, C., and Scheines, R. (2000). *Causation, Prediction, and Search*, second edition. MIT Press.

Tetrad Project (2026). *Tetrad 7.6.10 release and FCI source*. Official CMU repository, accessed 26 July 2026.

Word, E., Johnston, J., Bain, H. P., Fulton, B. D., Zaharias, J. B., Achilles, C. M., Lintz, M. N., Folger, J., and Breda, C. (1990). *Student/Teacher Achievement Ratio (STAR): Tennessee's K–3 Class Size Study. Final Summary Report 1985–1990*. Tennessee State Department of Education.

Zhang, J. (2008). On the completeness of orientation rules for causal discovery in the presence of latent confounders and selection bias. *Artificial Intelligence*, 172(16–17), 1873–1896.

[1] This document is the standalone final report; every number is regenerated from committed artifacts by the listed scripts. Extended per-stage evidence tables and complete source locators are kept in the repository's full report, reports/fci_plus_star_report.tex.